

Praveen Ghimire

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Education

University of Illinois at Chicago – Bachelor of Science in Mechanical Engineering

Dec 2024

Affiliations: AIAA, Engineering Design Team (EDT), UIC Robotics Lab

Projects

Robotic Arm with Pneumatic Grasping System

- Designed and built a robotic hand using a pneumatic system, servo motors, and embedded sensors for multi-object manipulation.
- Programmed control algorithms in C++ on Arduino hardware, reducing system response time from 2.0 s to 0.48 s.
- Integrated object detection and closed-loop feedback systems to enhance precision and automation.

Obstacle-Navigating Robot

- Designed and programmed a fully virtual autonomous robot in CoppeliaSim, navigating obstacle courses using simulated ultrasonic sensors and motor controllers.
- Developed real-time path-planning and collision-avoidance algorithms, enabling 100% completion of obstacle courses during testing.
- Validated robot performance entirely through simulation, demonstrating control system design, robotics programming, and virtual prototyping skills.

Motion-Sensor-Activated Showcase Box

- Developed and prototyped an automated showcase box with Arduino microcontrollers, ultrasonic sensors, and servo motors.
- Programmed motion logic for precise, touchless activation and fabricated parts using FDM 3D printing.

Experience

Mechanical Technician, Lakeside International Auto– Chicago, IL

Aug 2025 – Present

- Performed diagnostics and repairs on 10+ high-performance German vehicles weekly (Audi, BMW, Mercedes), addressing issues in powertrain, braking, suspension, and electrical systems.
- Utilized diagnostic scanners, oscilloscopes, and multimeters to pinpoint complex mechanical and electrical faults, reducing troubleshooting time by 25%.
- Implemented OEM-standard maintenance programs, cutting repeat service incidents by 20% and improving long-term vehicle reliability.
- Collaborated with technician teams to optimize workflows and documentation, enhancing service efficiency and customer satisfaction.

Field Engineer Intern, Eurotech Nepal – Kathmandu, Nepal

May 2022 – July 2022

- Audited reinforced concrete structures for compliance with Nepal National Building Code (NBC 105, NBC 202) and Nepal Standards (NS).
- Identified 25+ structural deviations and ensured corrective action to meet code and safety requirements.
- Prepared QA/QC and progress reports in Excel to support project supervision and documentation.

Founder & Head Internet Technician, Pooja Mobile and Communication – Kapilvastu, Nepal

Jan 2014 – March 2017

- Founded and scaled a wireless internet business serving 50+ households across remote communities
- Installed and configured 100+ routers and modems, ensuring stable coverage over a 10 km radius.
- Diagnosed and resolved 90% of service outages in the first visit, maintaining a 98% customer retention rate.
- Trained and supervised a 4-member team, reducing installation time per site by 30% through standardized procedures.

Skills

Engineering & Analysis: Machine Design, Thermal Analysis, Vibration Analysis, Control Systems, Prototyping, Diagnostics

Software & Simulation: SolidWorks, AutoCAD, ANSYS, MATLAB, Arduino, Python

Tools & Equipment: Multimeters, Oscilloscopes, Diagnostic Scanners, Hand & Power Tools, Soldering, Welding

Documentation: Technical Reporting, Engineering Drawings, Microsoft Office